

Swaragram based Residual Neural Architecture for Raag Identification in Indian Classical Music

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Abstract—Raag is the root of Indian Classical Music (ICM) in its arrangement, performance, improvisation, and organization. Applications of automatic raag identification include recommendation and indexing of ICM. This paper establishes a residual neural network (ResNets) approach that can differentiate a Raag from the swaragram representation of a song instead of previous methods on this task, which heavily depend on shallow machine learning. ResNets are very deep neural architectures that deal with the vanishing gradient issue and are usually used in the visual domain. Experiments were conducted on two standard datasets (Carnatic and Hindustani) with 40 and 30 Raags. The model achieved 79.9 % for the Carnatic and 90.4 % for the Hindustani dataset using the Swaragram. We have tested the model with many features and used the best feature to test against the state-of-the-art.

Index Terms—Indian classical music, Deep neural networks, Residual neural network, Raag identification

I. INTRODUCTION

Raag is a melodic structure of the Hindustan and Carnatic art music styles of the Indian subcontinent [1], [2]. Moreover, Raags are incorporated in various compositions of Indian folk and Bollywood music. It is a crucial component of ICM's exploration, organization, and teaching. An automated computer approach for Raag identification will allow retrieving music from comprehensive audio recordings, semantic-aware discovery and music navigation, and various music pedagogy applications. Here, we suggest an automated Raag identification solution in ICM audio collection.

A Raag is distinguished by a set of swaras (analogous to notes), Aaroh-Avroh (Ascending-Descending sequence of notes), and most notably by a set of melodic phrases (Pakad). These melodic phrases catch the essence of a Raag and constitute the composite and improvised building blocks of the ICM [1] melodies. They also act as leading indicators for Raag's identification by seasoned listeners [3].

While melodic phrases are essential for Raag, they are seldom used by computational approaches to identify Raag. The difficulties associated with accurate melodic pattern extraction from audio recordings of the ICM can be a reason for their non usage. These difficulties include the extraction of a consistent melodic representation, the absence of a musically significant paradigm of melodic segmentation, the difficulty in deriving melodic similarity metrics of perceptive relevance, and the

capacity to work with extensive audio collections. Study in computational modeling of ICM has been concentrated in recent years on overcoming these problems [4], [5].

In this article, we suggest using the Swaragram feature [6] in a novel method for Raag identification. To do this, we have used our earlier work on the melodious feature that uses the sound of a piece of music and uses the spectrum mapping (using the microtones of Shruti and the notations of pitch MIDI), and the pitch class binning is done to get a swara based distribution of music samples in ICM. We evaluate our approach on various representative Carnatic and Hindustani music datasets, which are open to the public. It is the most extensive data collection of this kind in terms of records, Raag number, and overall length. Since our approach is built on melodic features, it is easy to understand the extracted feature and its representation.

The article is organized in the following way; section II gives brief related works on this task of Raag identification. We present our methodology in section III, followed by the evaluation performed in section IV. Results and discussion of the evaluations are presented in section V, and finally, we conclude our contributions in section VI.

II. RELATED WORK

Raag identification is one of the most investigated tasks in ICM's content-based retrieval. Many approaches to solving this task use the tonal [7]–[13] pitch distribution (PD). However, the temporal elements of the melody, such as melodic progressions and pakad, are not seen as central aspects of Raags. In separating Raags based pakad [3], these features are even more critical. Several pieces of research resolve the shortcomings of PD-based approaches by modeling [14]–[18] temporal dynamics. The methods are usually used the sequential details using melodic progression templates [16], n-gram distributions [14], or hidden Markov models [17]. As far as we know, there are only two methods [19] and [15], which use Raag pakad specifically. However, these methods rely on manually annotated pakad. As musicians do not accept the same number of characteristic pakad, they are also subject to subjectivity. This often restricts the use of these techniques in even modest collections of music in the modern world.

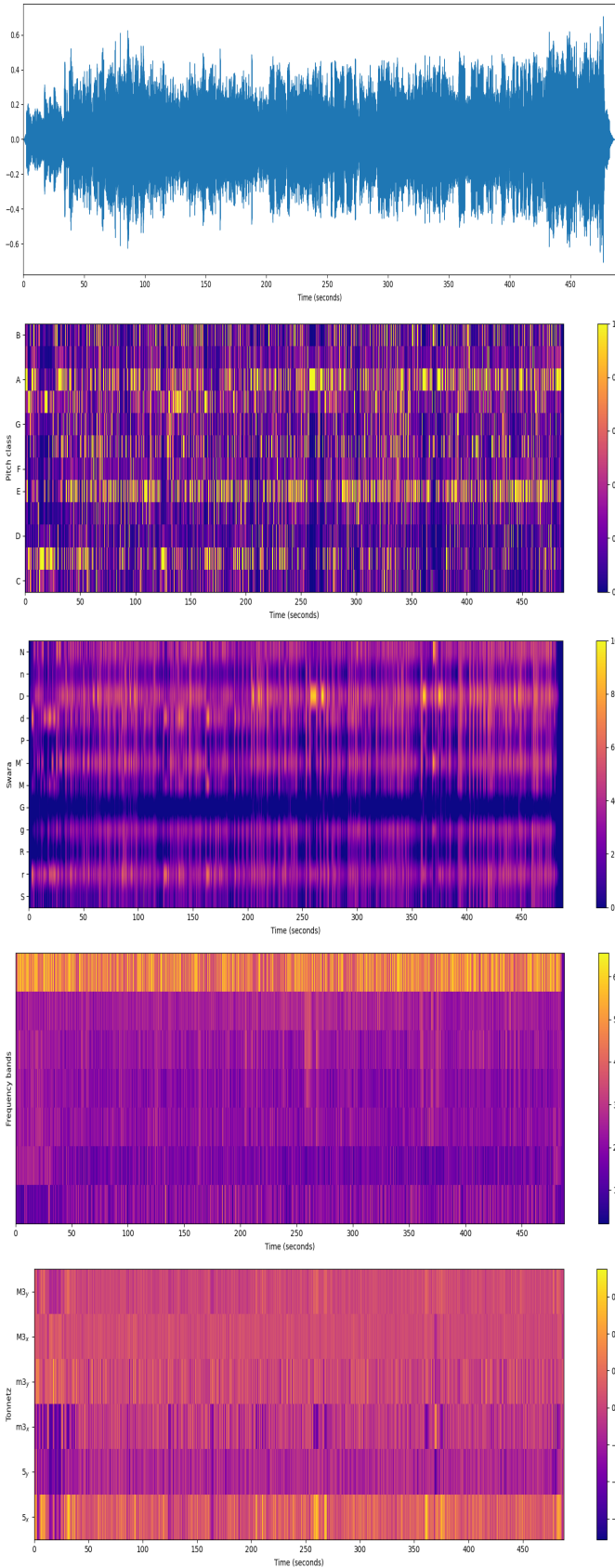


Fig. 1: (a) Waveform, (b) Chromagram, (c) Swaragram, (d) Spectral Contrast, and (e) Tonnetz visualization of Raag Bhoopali music sample

A sequence of swaras in a raga is one of the most common melodic features extracted computationally. Authors extract the component of swaras directly in an audio recording in [20] and [21]. This derived set of swaras is then combined to decide the song's raga underlying the available set of swaras for each raga. One failure of that kind of approach is that the timing aspects of the melody that are essential for identifying ragas are not taken into account. Many techniques are employed to capture the temporal facets of the [7], [11], [14]–[16], [18] melody using the Raag sequence Aaroh-Avroh. These methods often quantify the approximate pitch vectors to provide a distinct melody representation. Authors use quantization of notes and mapping of octaves in the [22] process to obtain the Aaroh-Avroh used for Raag identification. Such an unconscious representation will therefore produce errors, which impair the finished precision of the method.

Recent experiments on time characteristics have distanced themselves from the quantization of notes, and now the focus for extracting features is on the use of continuous pitch values. The authors in [23] provide a scalable solution to the Raag identifying problem using automatic detection of melodic patterns from "Pallavi" Lines in Carnatic music. They use sensitive locality hashing to calculate Raag's similarity and assign Raag based on the similarity measure ranking. In [24], authors derive an example of the "term frequency-inverse document frequency" used to describe the temporal characteristic of a vector space model in natural language processing. Authors use a "time-delayed melody surface" in [25] to catch both the tonal and temporal characteristics of a melody. While both [24] and [25] perform very well against separate Ragas, the detailed melodic subtleties needed to distinguish allied ragas cannot be registered.

III. METHODOLOGY

We use data-driven methodology; a set of features is extracted from audio files and fed into a residual neural network architecture (ResNet50V2). In this study, the musical features employed by us are Chromagram, Spectral Contrast, Swaragram, and Tonnetz. Each of them represents different characteristics of music.

A. Features

1) *Chromagram*: The chromagram [26] transforms the time-frequency characteristics of a sound into a temporarily different pitch precursor. This method analyzes music with pitches that can be categorized (into 12 classes) meaningfully and whose tuning is in the equal-tempered scale. This transition is focused on the perceptive capability of the auditory system, and many fascinating mathematical properties have been observed. The core trait of chroma characteristics is to catch harmonic and melodic musical features while at the same time making timbre and instrumental shifts robust. It is represented as shown in Fig 1(b); solfege notations (C-D-E-F-G-A-B) are represented in the Y-axis at each interval of time shown in the x-axis.

2) *Swaragram*: In ICM, the grammar of the Raag is based on the tonic and Shrutis (micro-tones) used in its formation. Tonic (Shadja-Sa) is the first Shruti in an octave, which is the root of all other notes (Re, Ga, Ma, Pa, Dha, Ni). All notes are based on Tonic; there is no Naad in Indian music without Tonic. Swaras in other octaves are binned into swara classes. This theory of tonic and Shrutis in ICM is the basis of swaragram's methodology [6]. Fig 1(c) shows a representation of the Swaragram of sample music; swaras notation (S-R-G-M-P-D-N) are represented in the Y-axis at each interval of time shown in the x-axis.

3) *Spectral Contrast*: The variation between spectral valleys and peaks results in spectral contrast [27]. We consider spectral peaks, valleys, and variations in each continuum of musical elements. Harmonic components of music equate approximately to the solid spectral summits, and most frequently, the spectral valley is a non-harmonic component. Thus, the harmonic and non-harmonic musical components represent the relative distribution of spectral contrast characteristics approximately. It is represented as shown in Fig 1(d); Frequency bands are represented in the Y-axis at each interval of time shown in the x-axis.

4) *Tonnetz*: This feature [28] maps 12-bin chroma vectors to the 6-D polytope interior; the vertices of this polytope are mapped with the pitch classes. Closer harmonic relationships such as fifths and thirds appear at small euclidian intervals. In order to develop a harmonic shift for the frame n , It quantifies the Euclidian distance between analytical frames $n+1$ and $n-1$. A detection peak marks the transfer from a harmoniously secure area to a different one. It is represented as shown in Fig 1(e); Harmonic relations are represented in the Y-axis at each interval of time shown in the x-axis.

B. Residual Block

After the first ImageNet 2012 competition won by a CNN architecture (AlexNet), each successful deep neural network architecture minimizing the error rate uses more deep layers. Increasing the number of hidden layers succeeds for fewer layers, but the vanishing/exploding gradient comes up, which is a general issue of deep learning as the number of layers increases. Increasing the number of layers makes the gradient 0 or too high. Thus, as we increase the number of layers, the training and test error rate often increases. A modern architecture called Residual Network, or ResNet [29] was proposed in 2015 by Microsoft Research scientists to overcome this problem. Fig 2 shows an architecture that implements the Residual Network's principles. This network solves the vanishing/exploding gradient problem. It uses a method called skip connections, these connection skips the training directly from several layers up and connects to the layers below.

The technique behind this network is that we allow the network to match the residual mapping instead of layer mapping. Therefore, let the network fit $\mathcal{F}(x)$ instead of $H(x)$, the original mapping $\mathcal{F}(x) := H(x) - x$ that gives $H(x) := \mathcal{F}(x) + x$.

The benefit of adding this skip connection is that if a layer damages the architecture's efficiency, regularisation will

overcome it. It leads to a very deep neural network formation without complications due to the vanishing/exploding gradient.

IV. EVALUATION

A. Data collection

Two music collections are used in this study, Carnatic and Hindustani. The music collections are collected from the CompMusic project [24], [25]. The Carnatic collection contains 124 hours of audio recordings of Carnatic music from 40 Carnatic Raags available on the market. There are 12 pieces for every Raag, representing a total of 480 recordings. At the same time, 116 hours of audio and editorial metadata, including the carefully selected and checked Raag labels, are included in the Hindustani collection. There are 300 recordings of 30 Raags and ten records of each Raag in the Hindustani collection.

In Musicbrainz [30], an open-access metadata archive, all editorial metadata for each audio recording is freely accessible. For the Carnatic and Hindustani, the musical collections compose primarily of 65 and 60 distinct singers. Collections belong to various Carnatic and Hindustani music forms, and there are 310 and 270 separate compositions. The Raags selected include various swaras (notes) sequences in both number and pitch groups (Swarasthan).

B. Experimental Setup

We performed the following preprocessing steps for each song:

- 1) Extract the features for every 1-sec segment as in [6]. This gives us a 12×44 feature matrix of Swaragram for 1 second (Perform similar step for other features [31]).
- 2) Slide sampling window by 1/2 second and repeat step 1 until the duration of the music sample.
- 3) Perform zero-padding if the last segment of the music sample is less than 1 sec in duration.
- 4) Perform a log normalization of the feature matrix.
- 5) Save the feature matrix as grayscale images for each segment along with its Raag label.

Every Raag in the Carnatic dataset has 12 songs and 10 songs in the Hindustani. Thus, for every Raag, we have enough feature dataset. The grayscale images of 12×44 (for Swaragram, different for other features) are fed to the residual

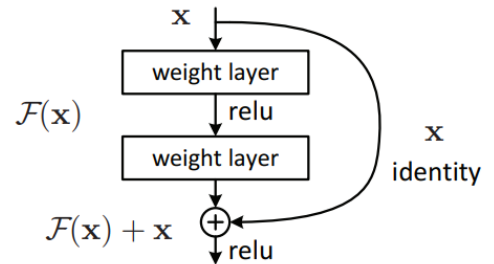


Fig. 2: Residual connection between layers

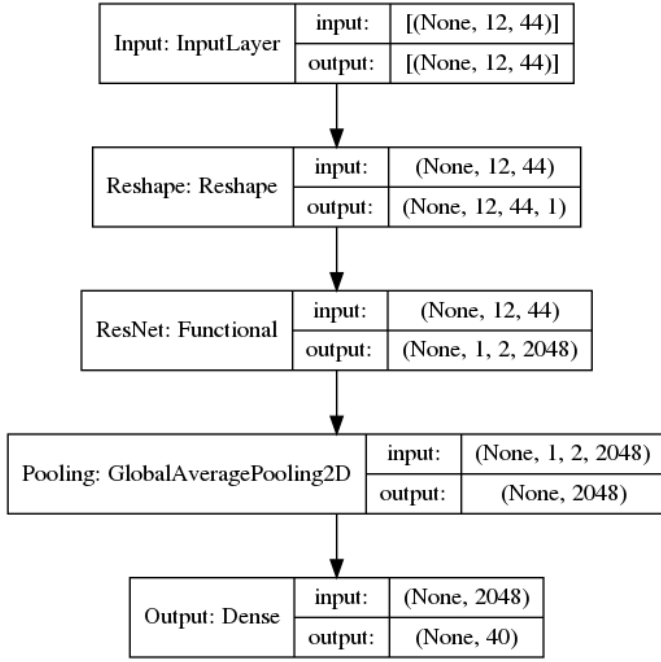


Fig. 3: Residual Neural network as a hidden layer of the employed model along with hyperparameter for Swaragram feature

neural network, which shows the variance of the Swaragram over time. We divided the data in the ratio of 80:10:10 in all data sets into trains, tests, and validation. We execute the model with different musical features. In addition, the following network parameters are used:

- ResNet50V2 residual neural network is utilized to overcome the vanishing gradient and better generalization of the data as shown in Fig 3.
- Global Average Pooling is performed on the output of the residual network.
- Training is performed using Adam optimizer [32] Soft-max activation is used in the last dense layer for the Sparse-Categorical-CrossEntropy loss function.
- Batch size of 64 is used along with 0.001 learning rate.
- Model's hyperparameters change depending on the musical feature employed.
- The model is trained for 100 epochs in each iteration of the experiment.

We use the Tensorflow v2.4.1 framework to execute our model. The experiment was conducted on Nvidia GPU Quadro RTX 5000, having 3,072 cores and 16 GB of dedicated memory.

C. Comparison with state-of-the-art

We equate our results with the three existing excellent methods [8], [13] and [23]. The system in [8] uses the tonal feature of the smooth pitch-class distribution and the nearest neighbor classifier using a distance of Bhattacharya for the Raag label prediction. We indicate this procedure by T1. In

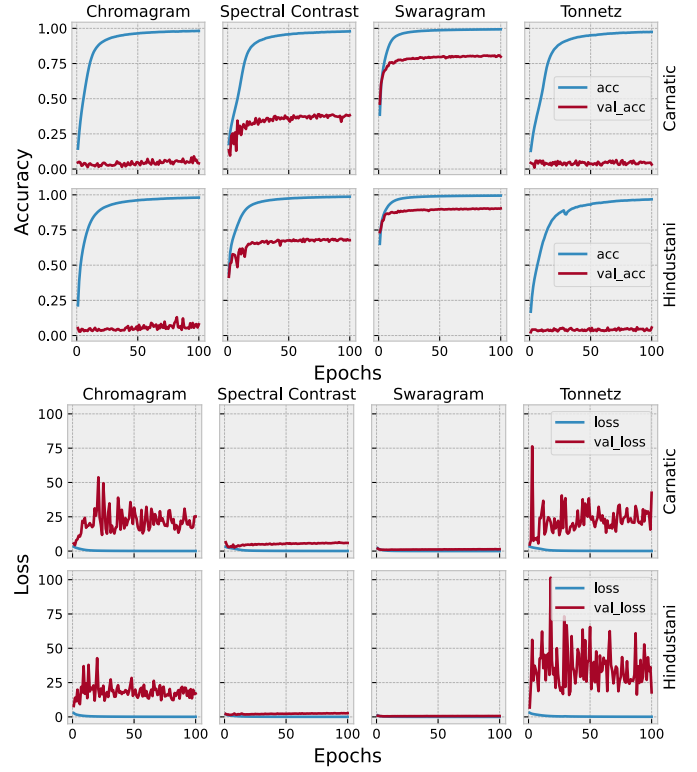


Fig. 4: (a) Accuracy of ResNet model for both dataset using different features. (b) Loss of ResNet Model for both dataset using different features.

[8], the authors claim a 120 secs segment size as the optimum analysis length. Notice that the authors do not experiment with a segment size less than 120 sec in [8].

In [13], the system also uses a pitch distribution-based feature. Unlike [8], however, the authors are using individual swaras as features of the parameterized pitch distribution. We indicate this procedure by T2. In [23], authors extract phrase-based features and use the Multinomial Naive-Bayes classification method to compute music samples' Raag category; we denote it as T3. All these methods conducted experiments on the Carnatic dataset with the original implementations of the methods described in [23]. In Table I, we denote our approach by Y.

V. RESULTS AND DISCUSSION

The findings of the proposed Y-method and the three state-of-the-art T1, T2, and T3 on two data sets, Carnatic and Hindustani, were presented in Table I. The highest accuracy is highlighted in bold for each dataset. Due to the insufficient findings on the Hindustani dataset for T1, T2, and T3 methods, only our Y method's results are shown. We begin by evaluating the effects of each approach on the Carnatic Dataset. The highest accuracy of Y for Carnatic is 79.9%, as shown in the table I. Compared to other approaches, the efficiency of any

Raag variant using Y has increased significantly, while 90.4% accuracy for the Hindustani set is achieved.

Furthermore, we see that the consistency of the ResNet Architecture over the other feature approaches is almost identical for both datasets in Table II, with no statistically significant gap. Given tonic information and dynamic frequency mapping dependent on Shruti, this is adequate for Raag identification. This result is interestingly consistent as tonic and shrutis employed are unique to a Raag, and it is enough to classify the Raag [6] from one occurrence of this pattern. The efficiency of the proposed method, as seen in Fig 4(a), is susceptible to the preference of deep learning architecture. We note that the highest accuracy with the Swaragram feature is achieved for both the datasets and their output relative to any other feature.

Note that in the image classification community, the ResNet classifier that exceeds other classifiers is well known. We, therefore, take the Swaragram as a comparison of our methodology with the ResNet classifier into account. Notice that the weights allocated by a classifier to the hidden layers may be used to classify the appropriate Raag Identification patterns.

We now review the training and validation phase of ResNet architecture, and we understand the errors in the classification of all features (Fig. 4b). We note that for both collections using the Swaragram, our approach achieves near-perfect accuracy. At the same time, for certain other features, we observe poor accuracy. When further exploring, ResNet architecture overwhelms within a few training epochs and begins remembering feature data instead of studying general patterns for these features. It is not easy to distinguish Raags based on overfitting because the distinctions are based on subtle melodic nuances. We also notice that many other features lack the adequacy to conform to the ICM.

In general, this classification error analysis suggests that our suggested approach is much more appropriate than other features to classify Raag. Further, This means that the approach based on the Swaragram can be mixed with other approaches to increase the accuracy of Raag identification.

TABLE I: Comparison of our approach Y with other states of the art.

Dataset	Models			
	T1	T2	T3	Y
Carnatic	74.1	51.4	69.6	79.9
Hindustani	-	-	-	90.4

TABLE II: Comparison of ResNet architecture over selected features. (C-Chromagram, SC- Spectral contrast, S-Swaragram and T-Tonnetz)

Dataset	Feature			
	C	SC	S	T
Carnatic	3.8	38.2	79.9	3.3
Hindustani	7.89	67.7	90.4	5.7

VI. CONCLUSION

We have proposed in this paper a new Swaragram based on ResNet architecture for Raag identification. None other approach uses a completely automatic mechanism to infuse Raag's tonic and complex microtones (Shruti) to the best of our knowledge. For 1 sec segments, few features are extracted with 1/2 sec sliding sampling. An architecture built on ResNet is used to classify Raags in two music sets. For assessment, we have assembled a significant list of representative features for both databases. The training results and the validation of classification with different features are shown in Fig 3. With these features, we have experimented and found that the Swaragram feature based approach is better than the others. Our findings suggest that audio recordings are suitable for identification because Raag has a simple presence or absence of tonic and dynamic microtones (Shruti).

Overall, we have shown that the Swaragram and ResNet-based method is a fruitful technique for detecting Raag and is superior to state-of-the-art methods. A review of the classification errors showed the complementary form of errors made using the phrase and Swaragram methods. In the future, we expect to explore the effective combination of both methodologies to enhance Raag detection.

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